

Teacher notes and answers

Resource use and carbon dioxide | LAUNCH_SU1_W01 | STAFF ONLY

Lesson alignment

Outcome: Explain resource use and climate change using online data.

Evidence: F: source annotation, evidence selection and questioning.

Teaching: Check source provenance, date and purpose. Annotate a small data set and identify a pattern; independently explain a link between resource use and climate change using evidence and a limitation.

Preparation

Prepare the cards, organiser and one task route. Model the four steps before independent work. Use the supplied source record and distinguish real evidence from authored scenarios. The teacher supplies the school's copy of No One Is Too Small by Greta Thunberg; no extract or quotation is included. Use the six NOAA rows and dry-air ppm units. Explain that the final selected year is the latest complete year in the downloaded file, 2025. Do not infer global temperature from one observatory.

Offer a calm start, an agreed pause and a private response option. Accept speech, pointing, typing or writing. Invite reasoned disagreement without personal or family disclosure.

Exact unit wording

Scheme of work: LAUNCH Summer 1

Unit

Resources climate and sustainability

Enquiry

How can we make a justified decision about sustainability?

Success criteria

I can interpret resource and climate evidence, evaluate responses, compare options against criteria and justify a practical decision.

Vocabulary

resource, sustainability, climate change, mitigation, adaptation, cause, effect, response, evidence, provenance, usefulness, decision, criteria

Resources

Climate and sustainability data; selected No One Is Too Small extracts by Thunberg; online data sets; curated sources; highlighters; prepared tables; spreadsheet or graph paper; planning frame; presentation device; glossary.

Access and challenge

Pre-teach terminology and model evidence use. Chunk sources and data; provide an initial frame, then reduce it. Maintain normal access arrangements. Extend by weighing limitations, alternative explanations and evidence quality. Adults support access and evidence selection without supplying answers. Use labelled examples and a reduced data set before the full task.

Teaching sequence

Nine stages totalling forty minutes

Stage	Min	Focus
Opening	0	State the enquiry and goal.
Arrival	4	Use supplied reminders and chosen route.
Starter	3	Use vocabulary in context.
I do	6	Model four steps before independent work.
We do	7	Operate the shared activity; explain choices.
Check	2	Check a claim against evidence.
Independent	12	Fade content prompts, retain access support.
Review	3	Check and improve one detail.
Exit	3	Hear two individual responses.

Worked model

Question: What pattern can I describe in the selected NOAA record?

Evidence: The annual means are 338.76 ppm in 1980 and 427.35 ppm in 2025.

Explain: These selected values rise. NASA links fossil-fuel burning to added CO₂ and warming.

Check: One observatory is not a global average. CO₂ concentration is not temperature.

Arrival answers

Staff only

Supported

1. NOAA Global Monitoring Laboratory.
2. ppm, parts per million, in dry air.
3. 2025 has 427.35 ppm.
4. No. It measures CO₂ concentration, not temperature.

Standard

1. Who made the source, when and why.
2. 1980 is 338.76 ppm and 2025 is 427.35 ppm.
3. Burning fossil fuels releases CO₂.
4. It is an authored example of an unattributed claim with no supporting method or evidence.

Stretch

1. It distinguishes when data were measured from when the file was released or read.
2. No. Unselected years must be checked separately.
3. No. A global average requires wider observations.
4. A suitable temperature record with its own provenance and coverage.

Checks and assessment

Retain the learner's own response and support record

Who made the source, when and why.

These selected values rise. NASA links fossil-fuel burning to added CO₂ and warming.

Year	CO ₂ ppm
1980	338.76
1990	354.45
2000	369.71
2010	390.10
2020	414.21
2025	427.35

Units: ppm (parts per million), dry air

NOAA GML annual mean CO₂ file, created 5 September 2026; accessed 21 September 2026.

The selected values rise. $427.35 - 338.76 = 88.59$ ppm. This selected record is not a global average or temperature series. A blank means missing data. The retained graph blocks a new plot for empty value cells and asks for a number. All-zero input plots without an error; distinguish a checked zero from missing evidence.

Accept accurate pointing, speech, drawing, typing or writing. Require a named detail and reason where asked. Do not treat a copied model as an independent response.

Record the learner's own reasoning and support used. Keep the conclusion individual and use the exit to plan one next step.

Sources and evidence record

Keep with the teacher copy

Evidence and factual claim record: ../../Sources_and_checks.html#w01

Exact unit scheme and access text: ../../Sources_and_checks.html#sow

Authored scenarios and limits: ../../Sources_and_checks.html#authored

Full claims, URLs, access dates and reading results: root Sources_and_checks.html. New pictures are authored teaching diagrams. The teacher supplies the school's copy of No One Is Too Small by Greta Thunberg. This pack includes no quotation or extract.

Preparation and source checks

Learner's own response and selected evidence

Access support / content prompts / adult note

Accuracy check and next step

Lesson vocabulary

Use these definitions with the lesson and knowledge organiser.

provenance — Who made a source, when and why

concentration — Amount of a substance within a mixture

resource — Something people use to meet a need

limitation — A boundary on what evidence can establish